The Power of Scale: Internet-Scraped Training Data and Zero-Shot Learning

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University of Colorado Boulder
Spring 2026

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https://dannagurari.colorado.edu/course/neural-networks-and-deep-learning-spring-2026/

Review

- Last lecture
  - Revisiting learning: from solving tasks to building representations that generalize
  - The recipe: large-scale pretraining + lightweight fine-tuning
  - Multimodal test: can the recipe replace CNN-RNN solutions for vision-language tasks?
  - Speech test: can the recipe replace engineered solutions for speech recognition?
  - Vision test: can the recipe replace CNNs for image classification?
  - Programming tutorial
- Assignments (Canvas)
  - Lab assignment 3 (last one!) due in 1.5 weeks; we recommend starting ASAP... this one is involved!
- Questions?

Today’s Topics

- The quest for growth: the secret sauce for success
- Scaling laws: predictable gains from model size, data, & compute
- Fueling scale: harnessing the Internet as training data
- Revelation: few-shot and zero-shot abilities emerge
- Programming tutorial

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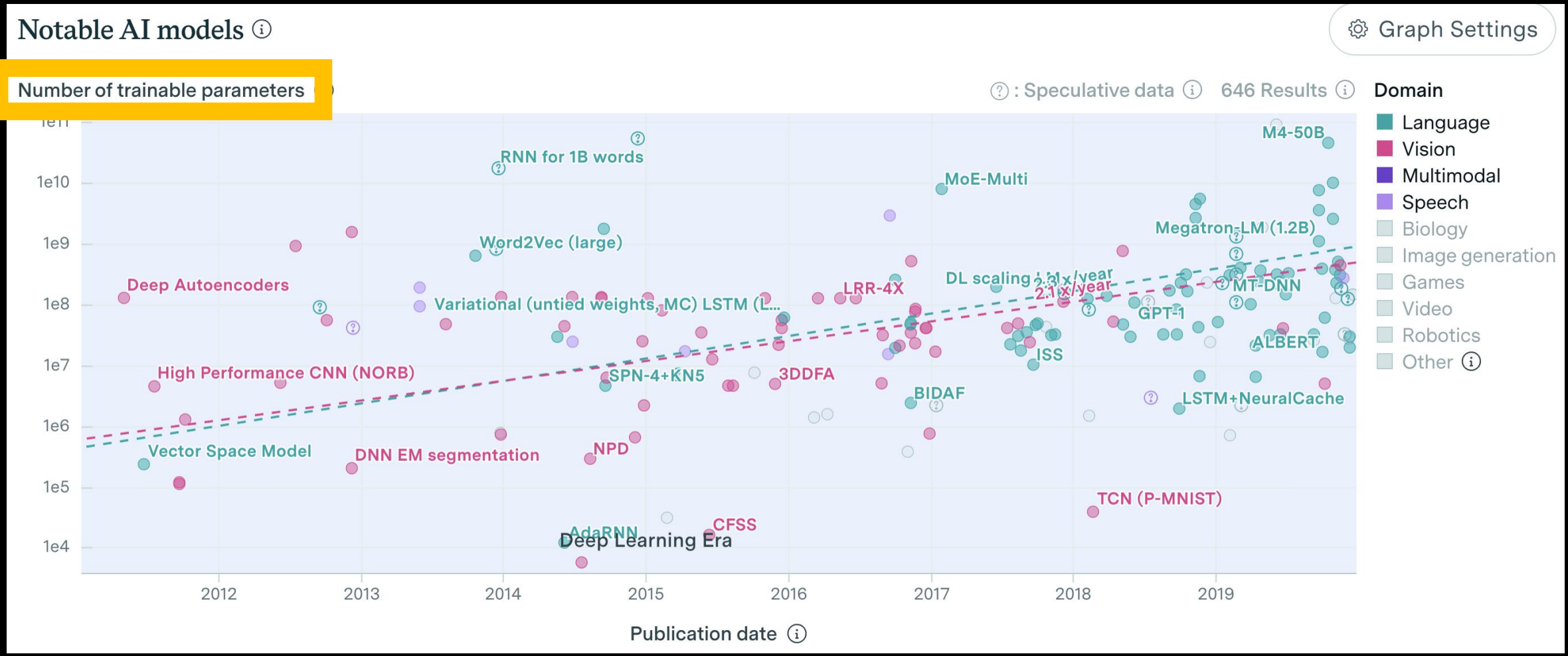
Context: Fine-Tuned Transformers Took Over



(Guiness World Record for most Transformers: 2,111)

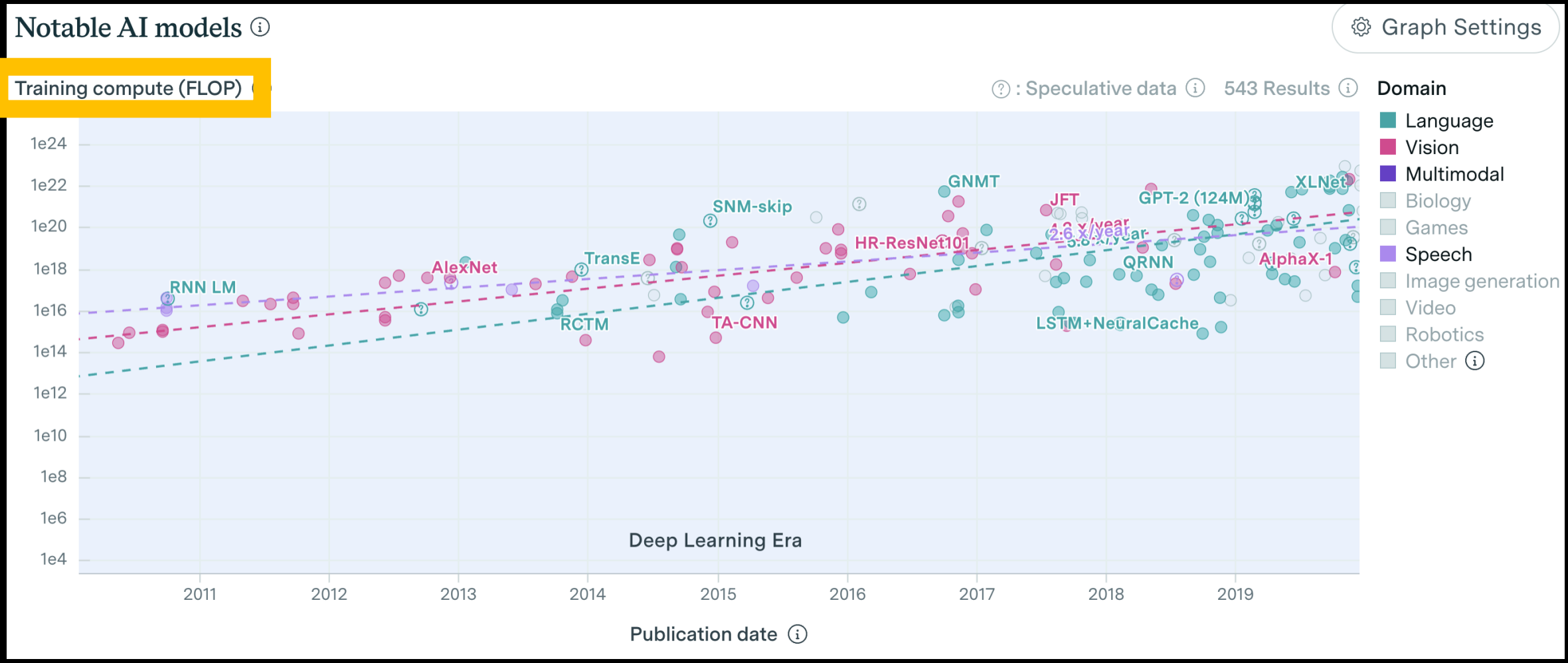
https://www.facebook.com/GuinnessWorldRecords/posts/the-largest-collection-of-transformers-memorabilia-is-2111-items-and-was-achieve/10157236275629032/

Trend: Bigger Model Was Secret Sauce for Achieving State-of-the-Art Performance



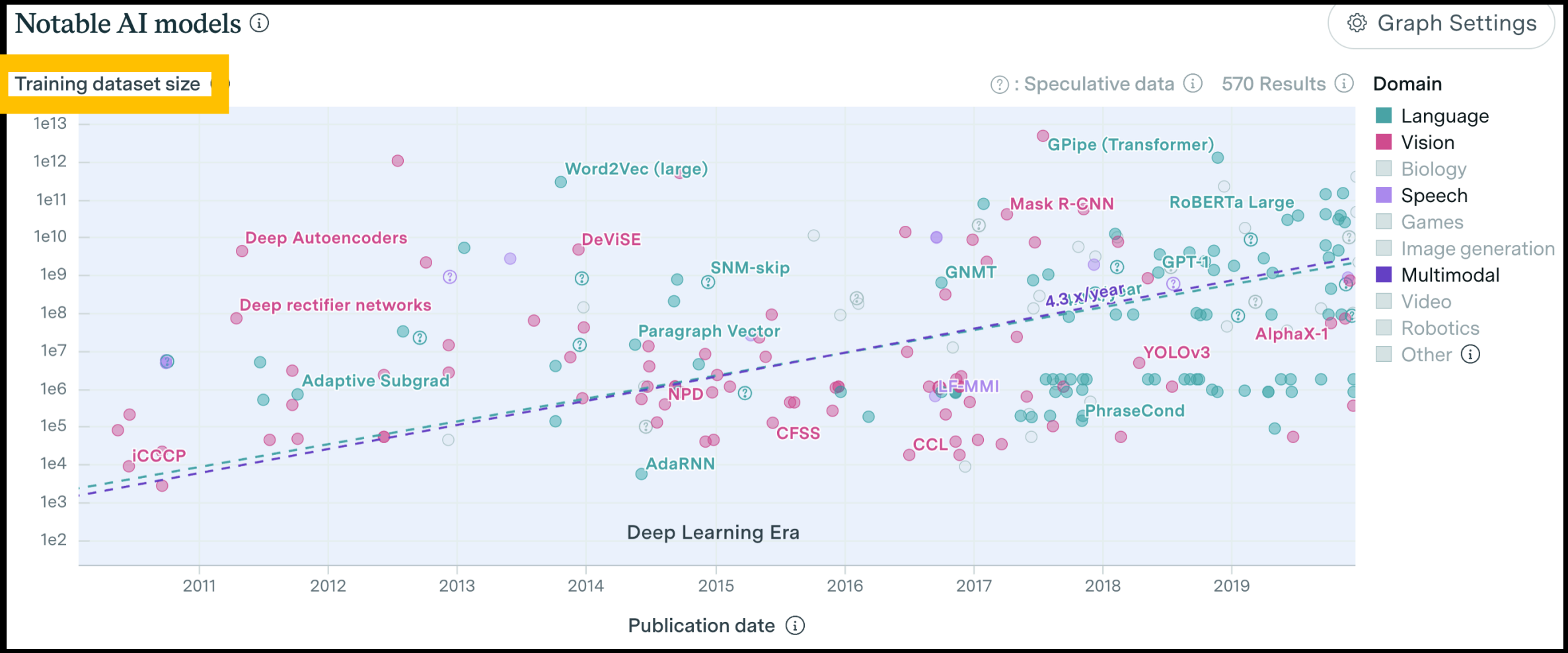
https://epoch.ai/data/ai-models

Empirical Evidence: Success Emerged with Sufficient Compute



https://epoch.ai/data/ai-models

Empirical Evidence: Success Emerged with Sufficient Compute and Training Data



https://epoch.ai/data/ai-models

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(arXiv 2017)

Deep Learning Scaling is Predictable, Empirically

Joel Hestness, Sharan Narang, Newsha Ardalani, Gregory Diamos, Heewoo Jun, Hassan Kianinejad, Md. Mostofa Ali Patwary, Yang Yang, Yanqi Zhou

"We would like to better analyze the relationships between training set scale, computational scale, and model accuracy improvements… [we] estimate scaling relationships for six deep neural network models across four application domains: machine translation, language modeling, image classification, and speech recognition Our results show that power-law learning curves exist across all tested domains. Although different applications yield different power-law exponents and intercepts, these learning curves exist across a broad range of models, optimizers, regularizers, and loss functions."

(arXiv 2020)

Scaling Laws for Neural Language Models

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"… language modeling performance improves smoothly and predictably as we appropriately scale up model size, data, and compute… The test loss of a Transformer trained to autoregressively model language can be predicted using a power-law…"

Background: Power-Law

output input

$$y = Cx^{\alpha}$$

constant

e.g., what happens to y if …

- $\alpha = 0.5$?

$$y = Cx^{0.5} = C\sqrt{x}$$

- If x increases by 4-fold, y increases by how much?

2

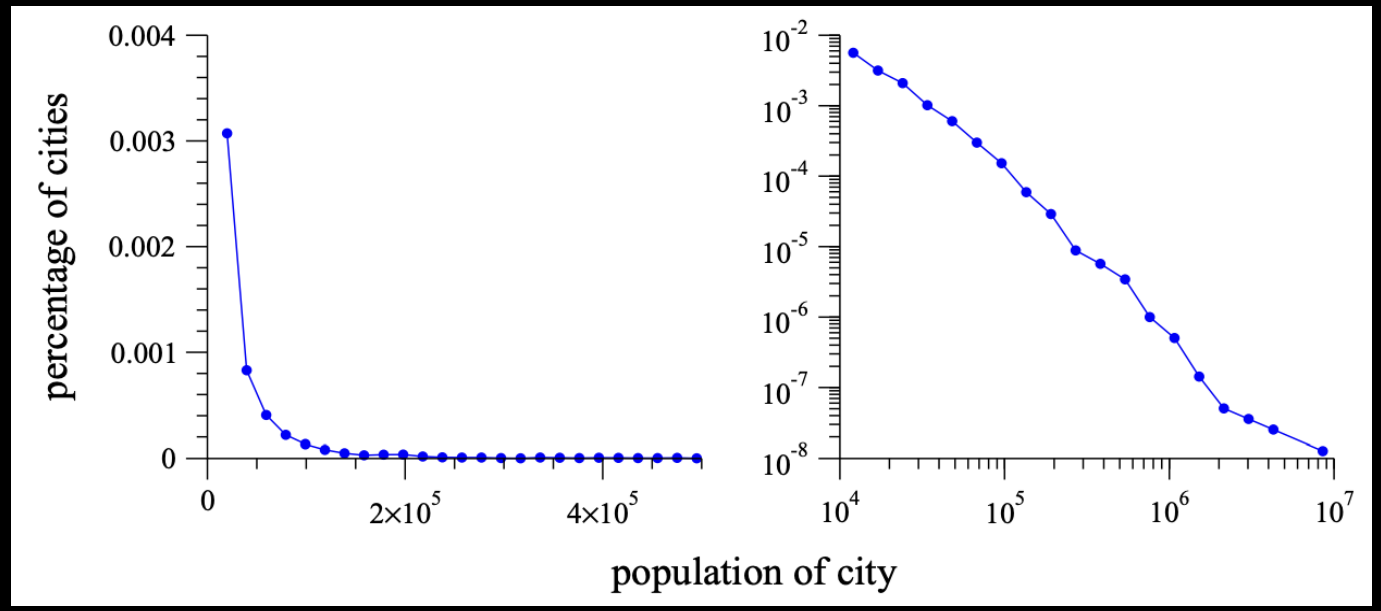
- More generally, for $0 < \alpha < 1$, scaling up the input yields predictable (smaller) gains for the output
  - How would you feel if this relationship held for the number of hours you put into studying versus your resulting grade?

Background: Power-Law (Negative Exponent)

output ↘ y = $Cx^{-\alpha}$ ↓ input

constant ↑ C

e.g., what happens when visualizing the original data instead on a log-log scale?



Newman. "Power laws, Pareto distributions and Zipf's law." Contemporary Physics, 2005.

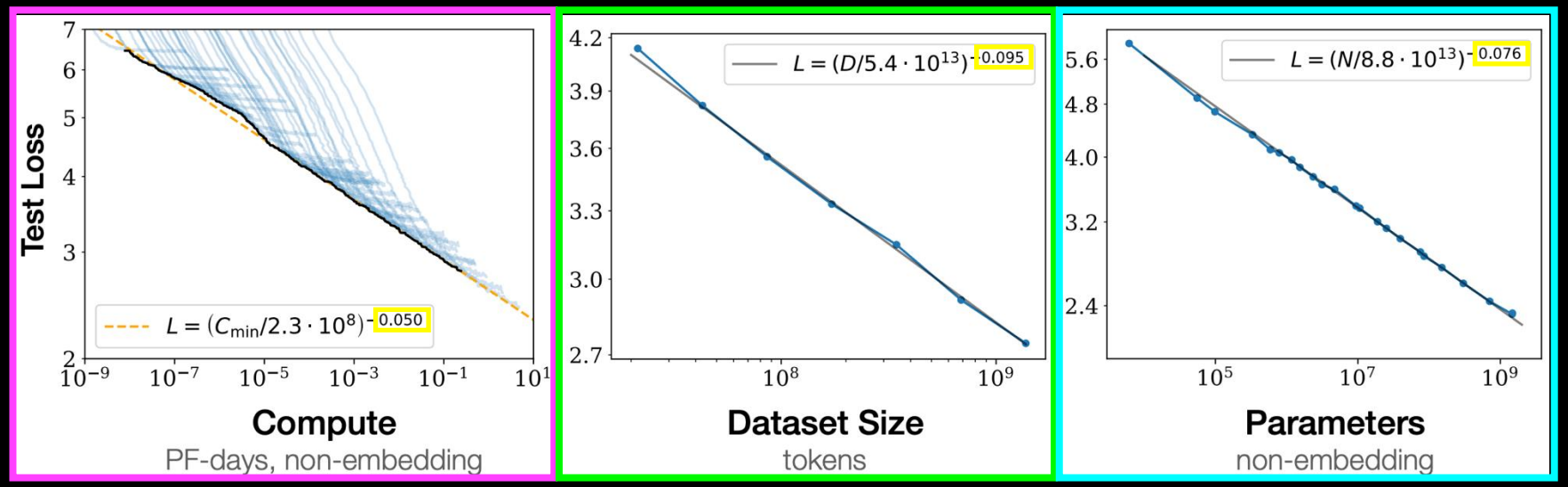
We get a straight line, with slope equal to $-\alpha$!

$(\log y = \log C - \alpha \log x)$

Empirical Results for Neural Language Models Autoregressively Predicting Next Token

- What law describes how test loss (L) is impacted by only changing the scale of:
  - Compute: C
  - Training data: D
  - Model parameters: N
- Power law relationships (straight lines on log-log scales)
  - What are the exponent values? Recall: $y = Cx^{-\alpha}$

Log-log scales:

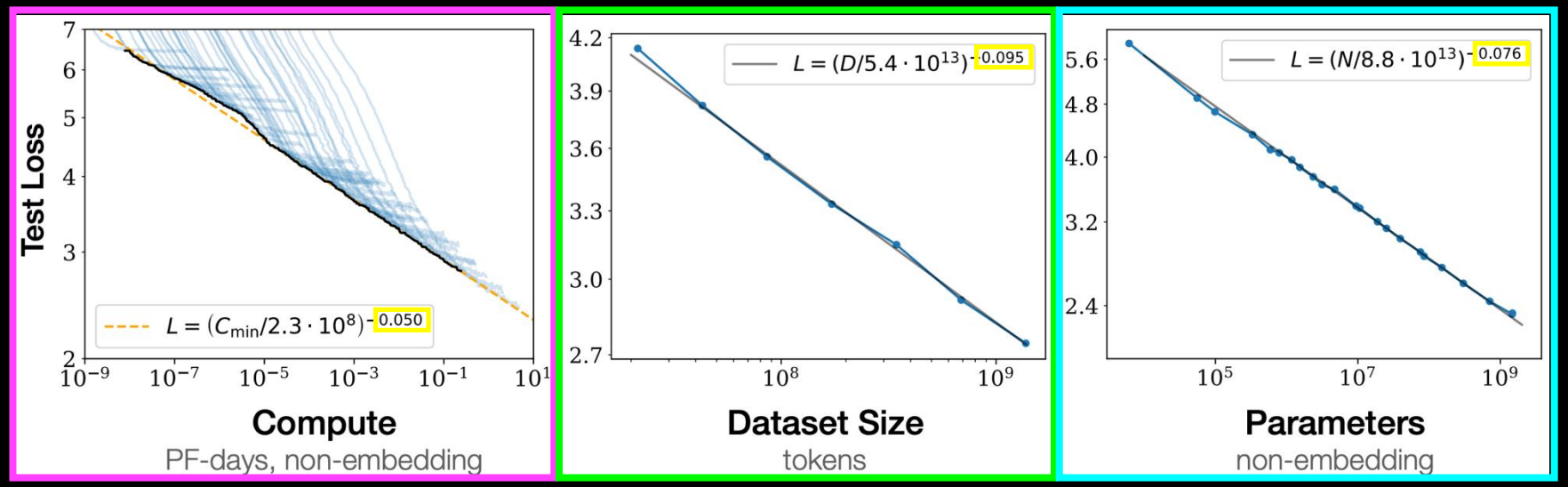


Kaplan et al. Scaling Laws for Neural Language Models. arXiv 2020

Empirical Results for Neural Language Models Autoregressively Predicting Next Token

- Up to what scale was tested for each factor…
  - Compute amount? – 10 PetaFLOP days = 86.4 x 10^{19} floating point operations
  - Training data size? – ~1B tokens
  - Number of model parameters? – ~1B
- What are future implications from the discovered power law relationship between test loss and scale of compute, training data, and model size?

Log-log scales:



Kaplan et al. Scaling Laws for Neural Language Models. arXiv 2020

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- Scaling with confidence: we can improve performance predictably by *just* scaling up compute, training data, and model size!

Kaplan et al. Scaling Laws for Neural Language Models. arXiv 2020

Subsequent Work Showed Scaling Laws Apply Across Many Domains and Tasks

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(GPT-2; OpenAI Blog 2019)

Language Models are Unsupervised Multitask Learners

Alec Radford [*1] Jeffrey Wu [*1] Rewon Child [1] David Luan [1] Dario Amodei [**1] Ilya Sutskever [**1]

(GPT-3; Neurips 2020)

Language Models are Few-Shot Learners

Tom B. Brown* Benjamin Mann* Nick Ryder* Melanie Subbiah*

Jared Kaplan† Prafulla Dhariwal Arvind Neelakantan Pranav Shyam Girish Sastry

Amanda Askell Sandhini Agarwal Ariel Herbert-Voss Gretchen Krueger Tom Henighan

Rewon Child Aditya Ramesh Daniel M. Ziegler Jeffrey Wu Clemens Winter

Christopher Hesse Mark Chen Eric Sigler Mateusz Litwin Scott Gray

Benjamin Chess Jack Clark Christopher Berner

Sam McCandlish Alec Radford Ilya Sutskever Dario Amodei

OpenAI

Architecture: Scaled-Up Sizes

GPT-2: minor modifications to GPT; e.g.,

- More layers (i.e., 48 vs 12)
- More parameters (1.5B vs 124M*)
- Layer norm layers re-positioned and added
- Parameter initialization modified
- Vocabulary expanded to 50,257 tokens
- Accepts up to 1,024 tokens instead of 512

GPT-3: minor modifications to GPT-2; e.g.,

- More layers (96 vs 48)
- More parameters (175B vs 1.5B)
- Accepts up to 2,048 input tokens instead of 1,024; is this equivalent to roughly:
  (a) 1/3 single-space page
  (b) 1 single-space page
  (c) 3 single-space pages

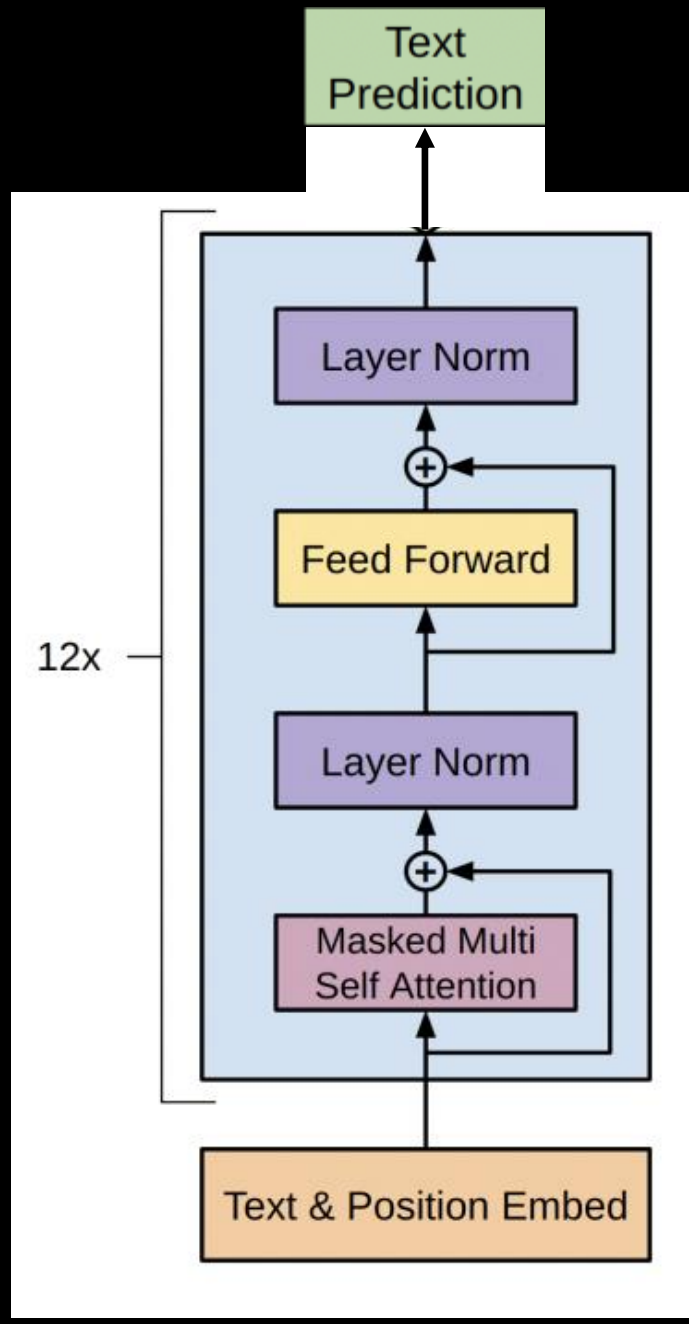


Image source: Radford et al. Technical Report 2018.

* Incorrect # reported in paper

Training Protocol

Same approach and objective as GPT (predict next word)

GPT-3: What do you think the training cost was?

(a) ~$5,00

(b) ~$50,000

(c) ~$500,000

(d) ~$5,000,000

* Required ~1.3 Gigawatt-hours of electricity (enough to power 121 homes in America for a year)

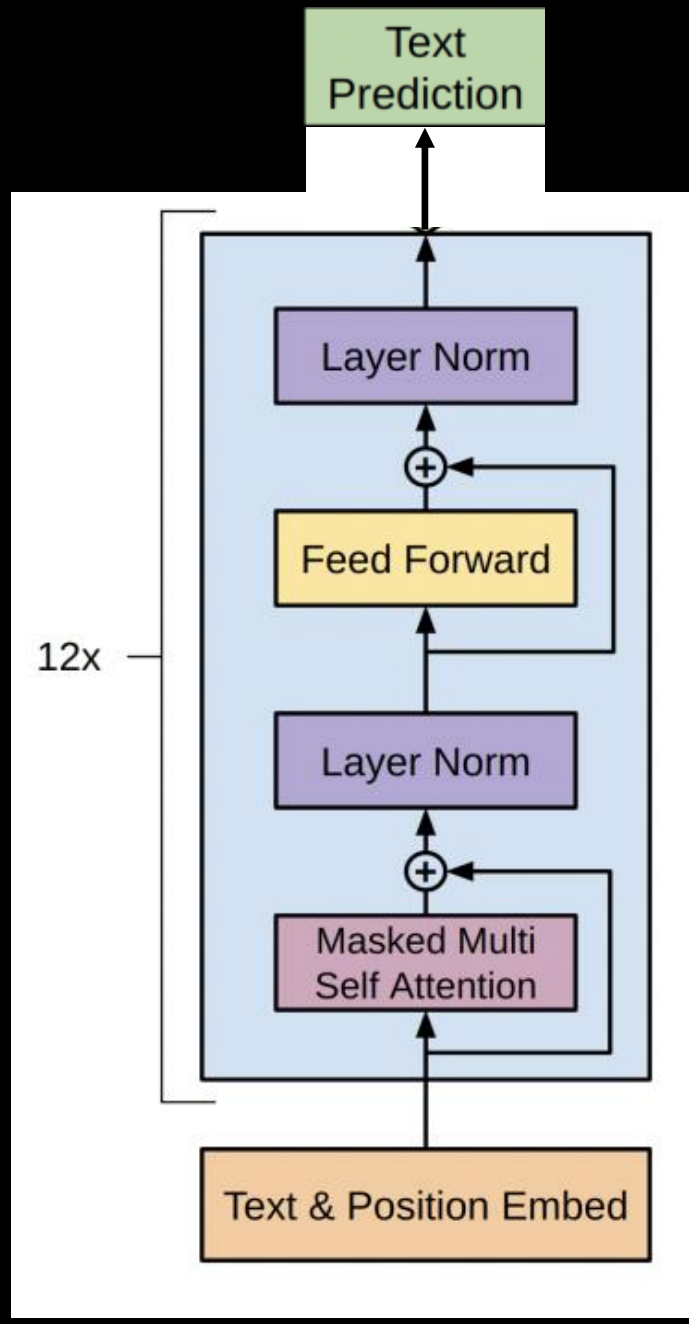


Image source: Radford et al. Technical Report 2018.

Training Data: Internet-Scale (Novelty)

Idea: natural language on the Internet represents many domains and tasks

GPT-2: WebText

Source: 

Quality control: “3 karma” rated outbound links that are de-duplicated and pre-processed

Result: 45M links through Dec 2017

Size: 8M documents = 40 GB

GPT-3: Multiple Datasets Including from Internet

Dataset	Quantity (tokens)	Weight in training mix
Common Crawl (filtered)	410 billion	60%
WebText2	19 billion	22%
Books1	12 billion	8%
Books2	55 billion	8%
Wikipedia	3 billion	3%

Noisy, lower quality Common Crawl data spanning 41 monthly Internet crawls from 2016 to 2019, supplemented with 4 high-quality datasets

How many GB? ~(a) 6, (b) 60, (c) 600, (d) 6,000



- Non-profit organization from California providing free monthly scrapes since 2008
- Web archives without HTML markup and non-text content https://commoncrawl.org/)
- Needs pre-processing to be more useful for training (e.g., spam, broken links, biased content)

Experimental Findings

GPT-2: strong performance on 7 of 8 tested NLP dataset challenges

- Their acknowledgments: “Thanks to everyone who wrote the text, shared the links, and upvoted the content in WebText. Many millions of people were involved in creating the data that GPT-2 was trained on.”

GPT-3: competitive and occasionally state-of-the-art performance for 10s of NLP dataset challenges

(CLIP; ICML 2021)

Learning Transferable Visual Models From Natural Language Supervision

Alec Radford [*1] Jong Wook Kim [*1] Chris Hallacy [1] Aditya Ramesh [1] Gabriel Goh [1] Sandhini Agarwal [1]
Girish Sastry [1] Amanda Askell [1] Pamela Mishkin [1] Jack Clark [1] Gretchen Krueger [1] Ilya Sutskever [1]

Novelty: image analysis models trained with natural language supervision using the vast amounts of publicly available data on the Internet

Architecture and Training Details

Task: predict which image-text pairs match

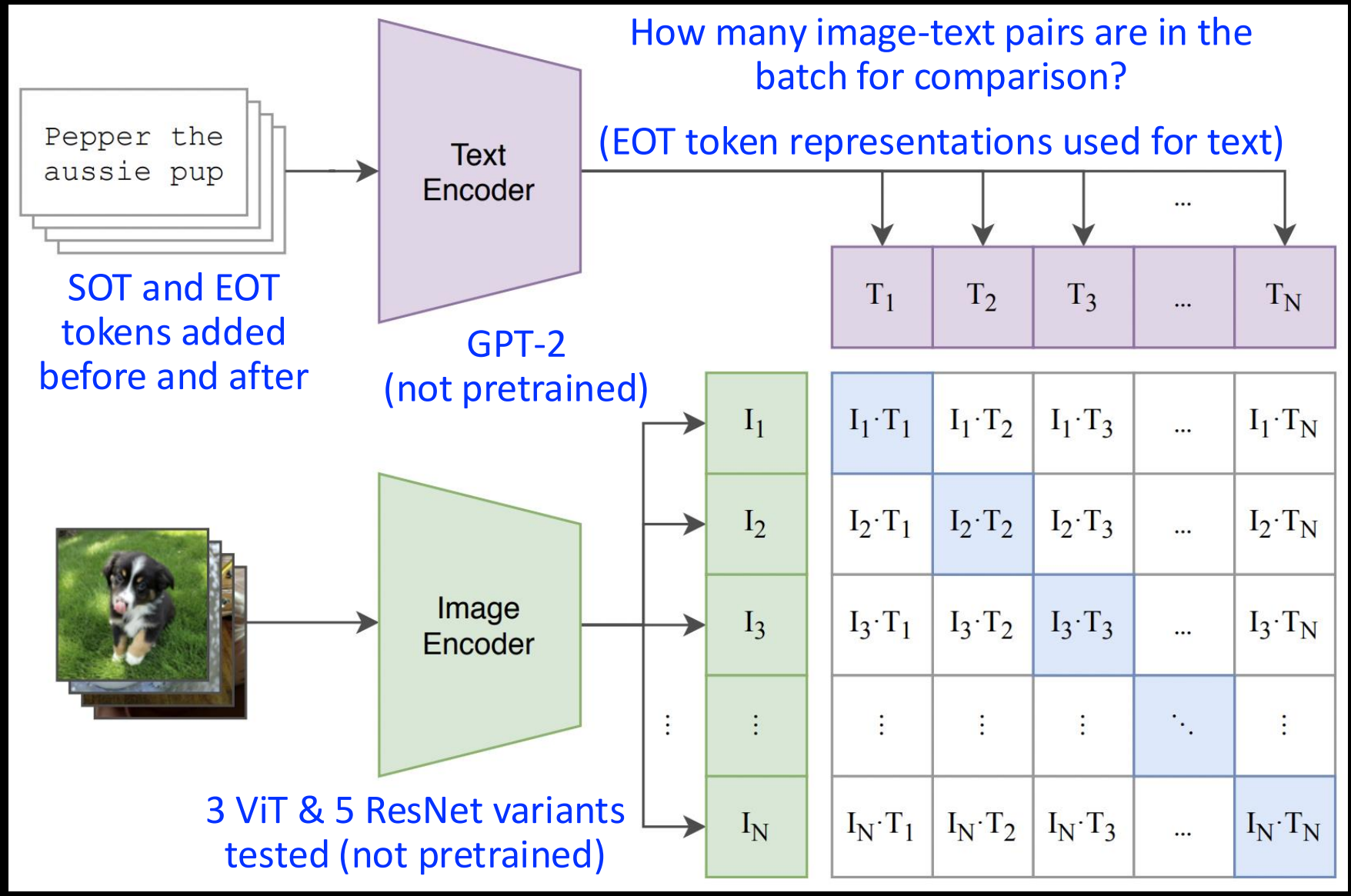
Data: 400 million image-text pairs from Internet

Model Size: ~100M to nearly 1B parameters across variants

Loss Function: sum of cross entropy for image and text classifications of their pairs

Optimizer: Adam

Training Duration: with largest ResNet, 18 days on 592 V100 GPUs; with largest ViT, 12 days on 256 V100 GPUs



Radford et al. Learning Transferable Visual Models From Natural Language Supervision. ICML 2021

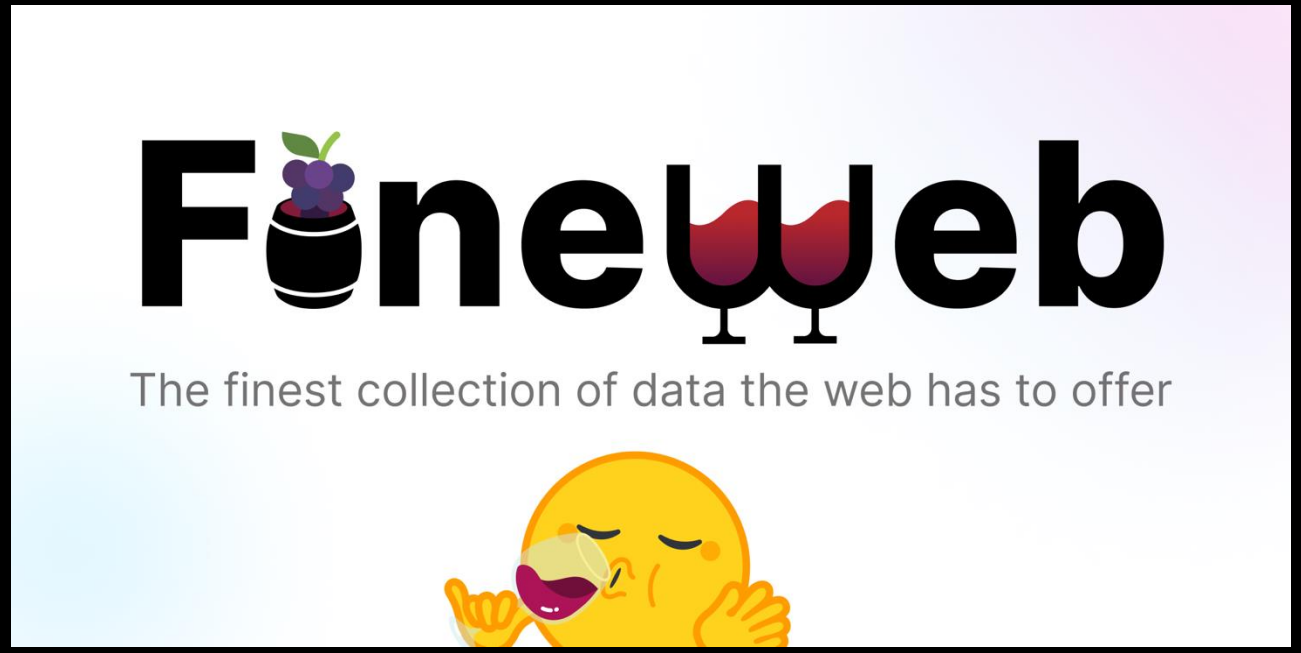
Experimental Findings: CLIP

Competitive performance on over 30 computer vision dataset challenges!

Many Internet-Scale Datasets Exist Today; e.g.,

	Tokens	Open source
SlimPajama	627B	Yes
RedPajama	1.21T	Yes
RefinedWeb-600B	600B	Yes
RefinedWeb-5T	5T	No
LLaMA	1.4T	No
MPT	1T	No
MassiveText	1.4T	No

https://cerebras.ai/blog/slimpajama-a-627b-token-cleaned-and-deduplicated-version-of-redpajama

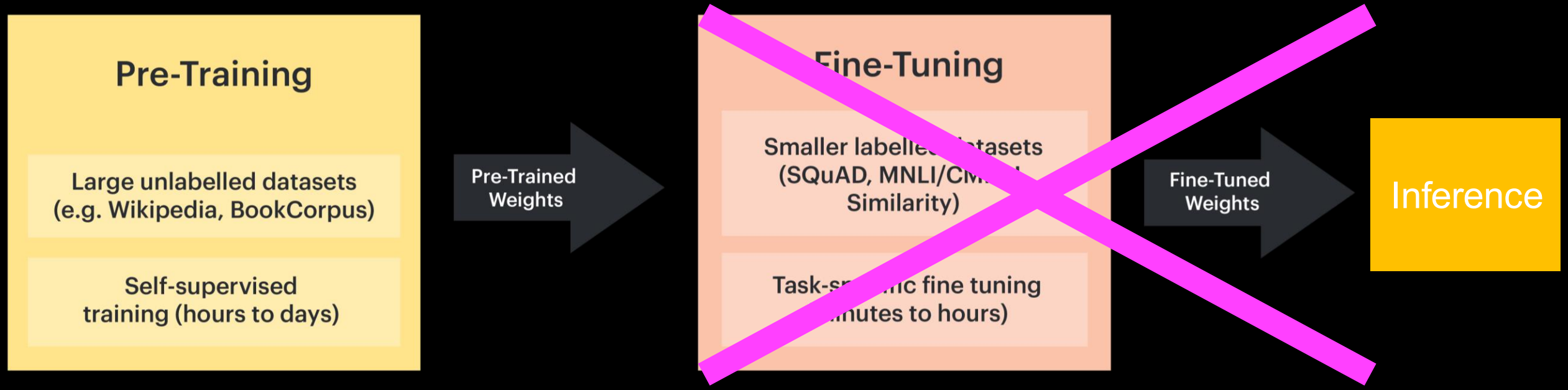


https://huggingface.co/datasets/HuggingFaceFW/fineweb

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Did You Notice? Pioneering Internet-Trained Models Did Not Center on Being State-of-the-Art

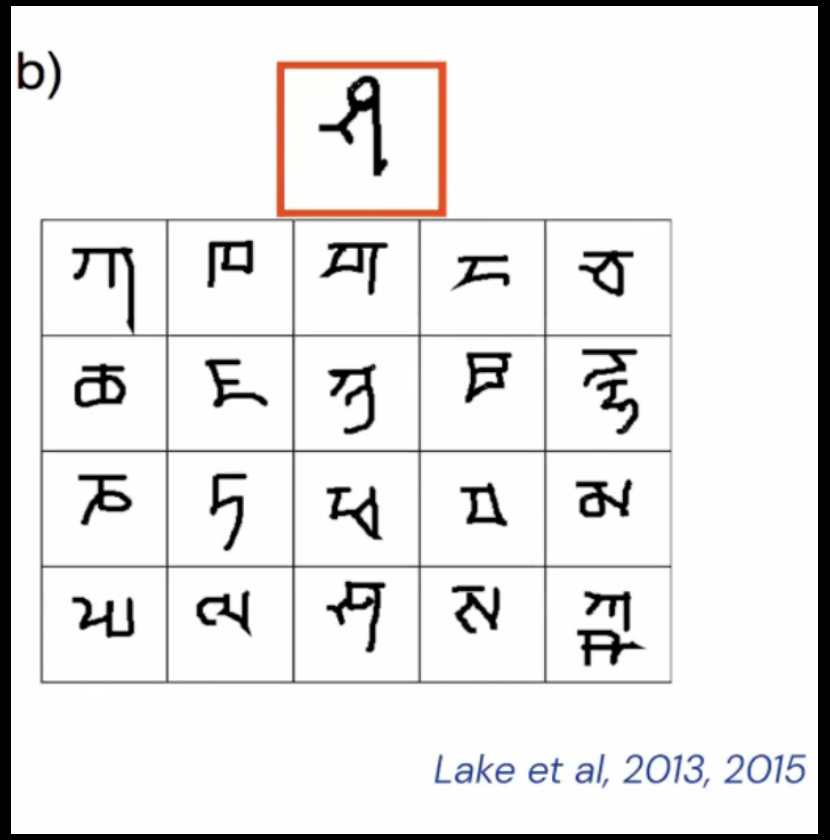


These models were important because they kicked off a paradigm shift to tackle downstream tasks in few-shot/zero-shot settings!

https://docs.graphcore.ai/projects/bert-training/en/latest/bert.html

Intuition: Can You Learn to Recognize Something With a Few Examples?





Given one example per category, identify the category of the query

https://www.youtube.com/watch?v=9j4iH9TPTd8

Intuition: How Many Examples Do You Need to See to Recognize What This Is?



0; just recognize it looks like a horse with black and white stripes

Intuition: How Many Examples Do You Need to See to Recognize What This Is?



0; just recognize it is the upper body of a person attached to the body and legs of a horse

Inference Time Input: Instructions + Optionally Examples (Latter Called "In-Context Learning")

Zero-shot

The model predicts the answer given only a natural language description of the task. No gradient updates are performed.

```
1  Translate English to French:        ← task description
2  cheese =>                           ← prompt
```

One-shot

In addition to the task description, the model sees a single example of the task. No gradient updates are performed.

```
1  Translate English to French:        ← task description
2  sea otter => loutre de mer          ← example
3  cheese =>                           ← prompt
```

Few-shot

In addition to the task description, the model sees a few examples of the task. No gradient updates are performed.

```
1  Translate English to French:        ← task description
2  sea otter => loutre de mer          ← examples
3  peppermint => menthe poivrée        ←
4  plush girafe => girafe peluche      ←
5  cheese =>                           ← prompt
```

Brown et al. Language Models are Few-Shot Learners. Neurips 2020.

GPT-2 Experimental Results: Zero-Shot Setting

	LAMBADA (PPL)	LAMBADA (ACC)	CBT-CN (ACC)	CBT-NE (ACC)	WikiText2 (PPL)	PTB (PPL)	enwik8 (BPB)	text8 (BPC)	WikiText103 (PPL)	1BW (PPL)
SOTA	99.8	59.23	85.7	82.3	39.14	46.54	0.99	1.08	18.3	**21.8**
GPT-1: 117M	**35.13**	45.99	**87.65**	**83.4**	**29.41**	65.85	1.16	1.17	37.50	75.20
BERT-sized: 345M	**15.60**	55.48	**92.35**	**87.1**	**22.76**	47.33	1.01	**1.06**	26.37	55.72
Larger: 762M	**10.87**	**60.12**	**93.45**	**88.0**	**19.93**	**40.31**	**0.97**	**1.02**	22.05	44.575
GPT-2: 1542M	**8.63**	**63.24**	**93.30**	**89.05**	**18.34**	**35.76**	**0.93**	**0.98**	**17.48**	42.16

Achieved state-of-the-art for zero-shot setting on 7 out of the 8 datasets

Radford et al. Language Models are Unsupervised Multitask Learners. OpenAI Blog 2019

Context (passage and previous question/answer pairs)

Tom goes everywhere with Catherine Green, a 54-year-old secretary. He moves around her office at work and goes shopping with her. "Most people don't seem to mind Tom," says Catherine, who thinks he is wonderful. "He's my fourth child," she says. She may think of him and treat him that way as her son. He moves around buying his food, paying his health bills and his taxes, but in fact Tom is a dog.

Catherine and Tom live in Sweden, a country where everyone is expected to lead an orderly life according to rules laid down by the government, which also provides a high level of care for its people. This level of care costs money.

People in Sweden pay taxes on everything, so aren't surprised to find that owning a dog means more taxes. Some people are paying as much as 500 Swedish kronor in taxes a year for the right to keep their dog, which is spent by the government on dog hospitals and sometimes medical treatment for a dog that falls ill. However, most such treatment is expensive, so owners often decide to offer health and even life _ for their dog.

In Sweden dog owners must pay for any damage their dog does. A Swedish Kennel Club official explains what this means: if your dog runs out on the road and gets hit by a passing car, you, as the owner, have to pay for any damage done to the car, even if your dog has been killed in the accident.

Q: How old is Catherine?
A: 54

Q: where does she live?
A:

Model answer: Stockholm
Turker answers: Sweden, Sweden, in Sweden, Sweden

Is a zero-shot or few-shot input used?

Context (WebText test)

Outfit 1: Typical

This pairing was the first outfit I thought of when I bought the shoes. It's like a summer version of this Jake Grantham outfit; in fact, my shoes are close to the colors of his Nike Racers! Instead of a heavy Harris Tweed jacket and denim shirt, I'm wearing a cotton DB jacket and and a linen shirt. Both fabrics (in these colors) are an absolute must for summer, as they go with both dark and and light pants! As you can see, they pair wonderfully with the dark jeans and shoes. It's a pseudo menswear/prep outfit.

Overall, this is a very casual outfit which is why I paired my sneakers with it. I'm not about wearing a full wool suit with sneakers (as GQ shows a lot) but I'm definitely open to keeping things casual, like this cotton DB. Casual fabrics are key to pulling off your sneakers in a dressed down menswear outfit. I'd even suggest to wear these sneakers with a khaki chino suit or a white linen suit. Just be sure to ditch the tie or wear a tee or polo; wearing a tie with sneakers is a bit too much

Small model completion

.

Outfit 2: High End

This outfit was just in the eye of the storm. It's a hybrid of what ASICS does with streetwear (like J. Crew) and what J. Crew does with casual pieces that are a bit more traditional. Again, as I mentioned before, it's a low budget item and they can be very expensive. These will definitely be an issue if you're trying to work as a casual sneakermaker, and that's OK but don't hesitate if you want to try the lower price versions of their outfits!

This version features a very low cut-cut neckline (which is not a bad thing), slightly longer legs, and a pair of long legs to keep things nice. My preferred color of color these items are blue-gray (which is my first colors, as they are a light white), or black (which is my favorite of my designs). As always, my pick of colors from this store have nothing to do with that other than the design. For a high end, I would recommend this to everyone!

Outfit 3: High Light

This was a "high light" look while I had a chance to get some color. I think the color is based on

GPT-2 completion

for me!

Outfit 2: Casual Jacket

I know that some people might be opposed to wearing sneakers in a turtleneck, but I wanted to be true to myself, so I went with a slim fitting turtleneck in a color more similar to my favorite color of the day. Overall, it's a classic turtleneck, with nothing too flashy. I did swap my shoes for these ASICS Tiger Killshots and I wish I hadn't did that. The shoe itself is very comfortable, however, I found that wearing them in these sneakers made my feet look a bit larger. I'd wear them with denim, chinos, whatever, but would never recommend wearing them alone. There are just too many variables involved.

Outfit 3: Dress

This is the final outfit I wore the ASICS Tiger Killshots. This version of the ASICS is a much more modern and streamlined version. The shoe has the most modern and minimal design that I can think of! With a lot of black, these sneakers look very cool. They'll definitely be very comfortable for summer.

To me, wearing them with a suit is the easiest way to have a good pair of sneaker shoes. That's pretty much it! If

Is a zero-shot or few-shot input used?

(predictions from smallest & largest models)

Radford et al. Language Models are Unsupervised Multitask Learners. OpenAI Blog 2019.

Few-shot format:

- english sentence = french sentence pair, followed by
- english sentence =

English reference	**GPT-2 French translation**
This re-release, titled The Next Day Extra, was presented in the form of three disks: the original album, unpublished studio sessions and remixes, plus a DVD containing the four clips that have already been unveiled.	Les nouvelles re-releases, tout en premier disc, nécessaire de l'album, un studio session et remixes, plus une DVD de l'écran de quelques clips qui ont été déjà échappés.
French reference	**GPT-2 English translation**
Un homme expliquait que le fonctionnement de la hernia fonctionnelle qu'il avait reconnaît avant de faire, le fonctionnement de la hernia fonctionnelle que j'ai réussi, j'ai réussi.	This release, originally released on three discs: the original album, sessions studio inédites and remixes, a DVD containing the four clips that were released on the previous day.

What does the model's success tell us about the Internet-scraped training data?

Radford et al. Language Models are Unsupervised Multitask Learners. OpenAI Blog 2019

GPT-3 Experimental Findings: Could Compete with Fine-Tuned State-of-the-Art Models; e.g.,

Setting	PTB
SOTA (Zero-Shot)	35.8[a]
GPT-3 Zero-Shot	**20.5**

Setting	LAMBADA (acc)	LAMBADA (ppl)	StoryCloze (acc)	HellaSwag (acc)
SOTA	68.0[a]	8.63[b]	**91.8**[c]	**85.6**[d]
GPT-3 Zero-Shot	**76.2**	**3.00**	83.2	78.9
GPT-3 One-Shot	**72.5**	**3.35**	84.7	78.1
GPT-3 Few-Shot	**86.4**	**1.92**	87.7	79.3

Setting	NaturalQS	WebQS	TriviaQA
RAG (Fine-tuned, Open-Domain) [LPP+20]	**44.5**	**45.5**	**68.0**
T5-11B+SSM (Fine-tuned, Closed-Book) [RRS20]	36.6	44.7	60.5
T5-11B (Fine-tuned, Closed-Book)	34.5	37.4	50.1
GPT-3 Zero-Shot	14.6	14.4	64.3
GPT-3 One-Shot	23.0	25.3	**68.0**
GPT-3 Few-Shot	29.9	41.5	**71.2**

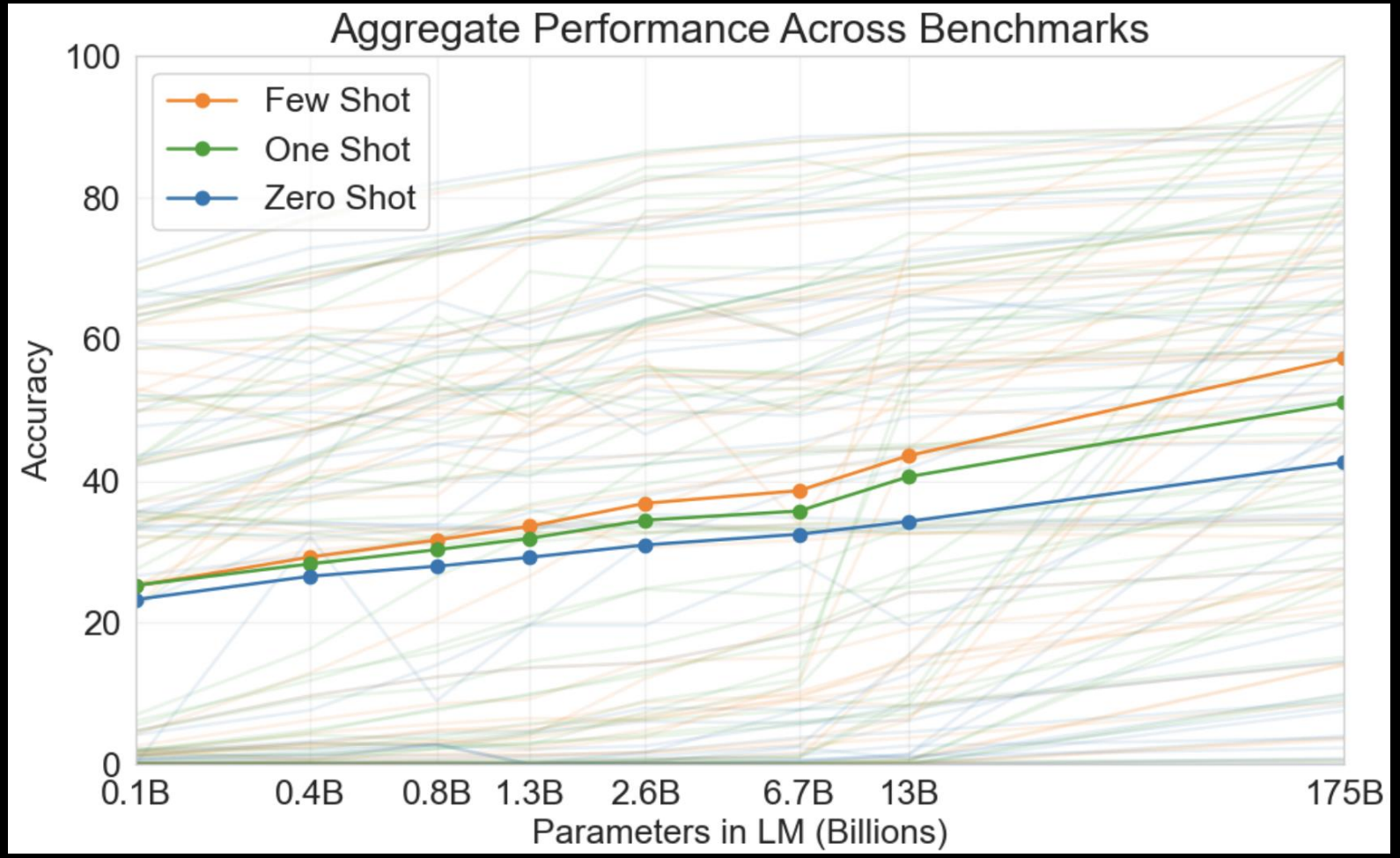
Brown et al. Language Models are Few-Shot Learners. Neurips 2020.

Group Discussion: Empirical Trends for GPT-3

How does performance change as we vary the number of in-context examples (0, 1, few)?

How does model size affect that relationship?

(42 NLP challenges)

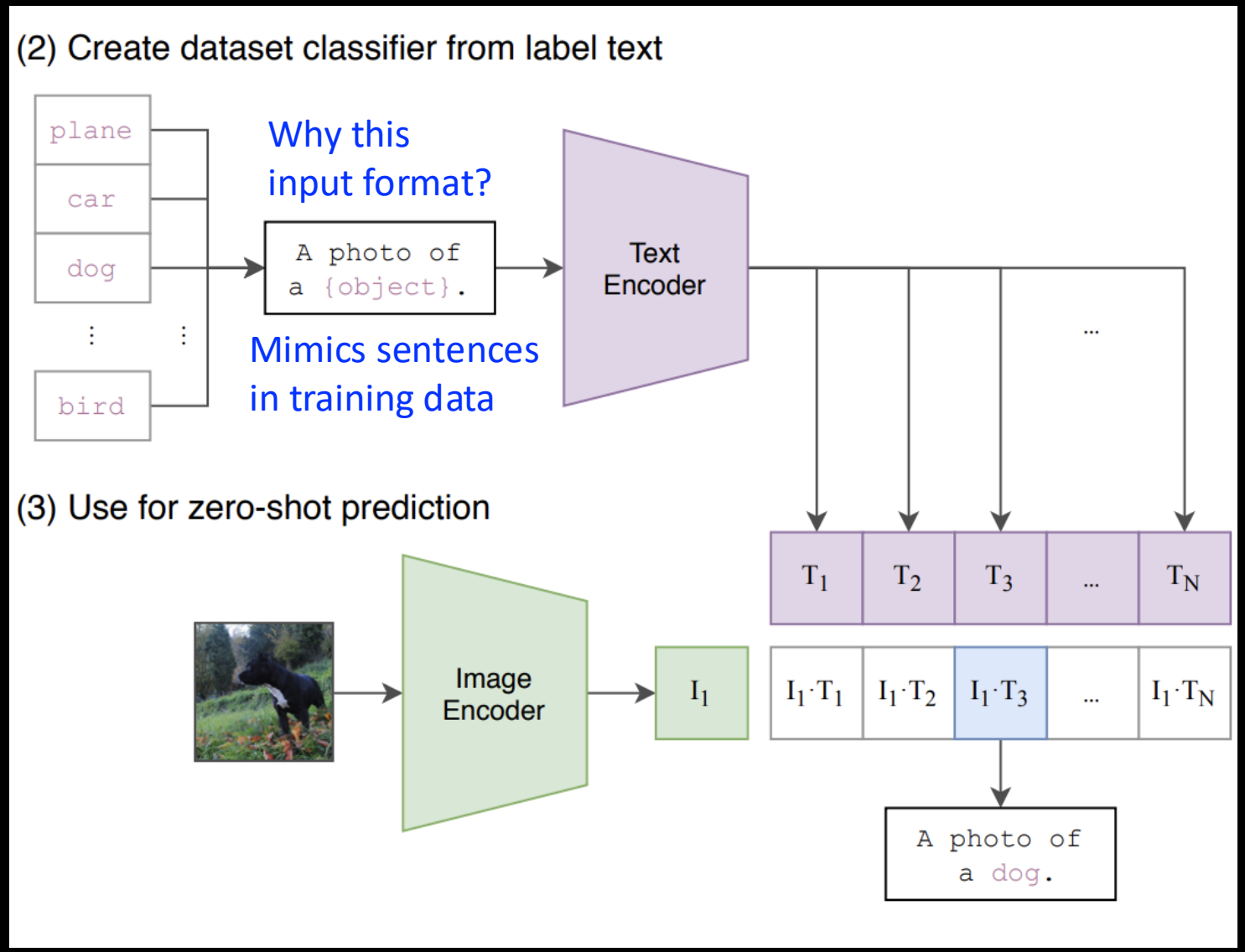


Brown et al. Language Models are Few-Shot Learners. Neurips 2020.

CLIP Inference

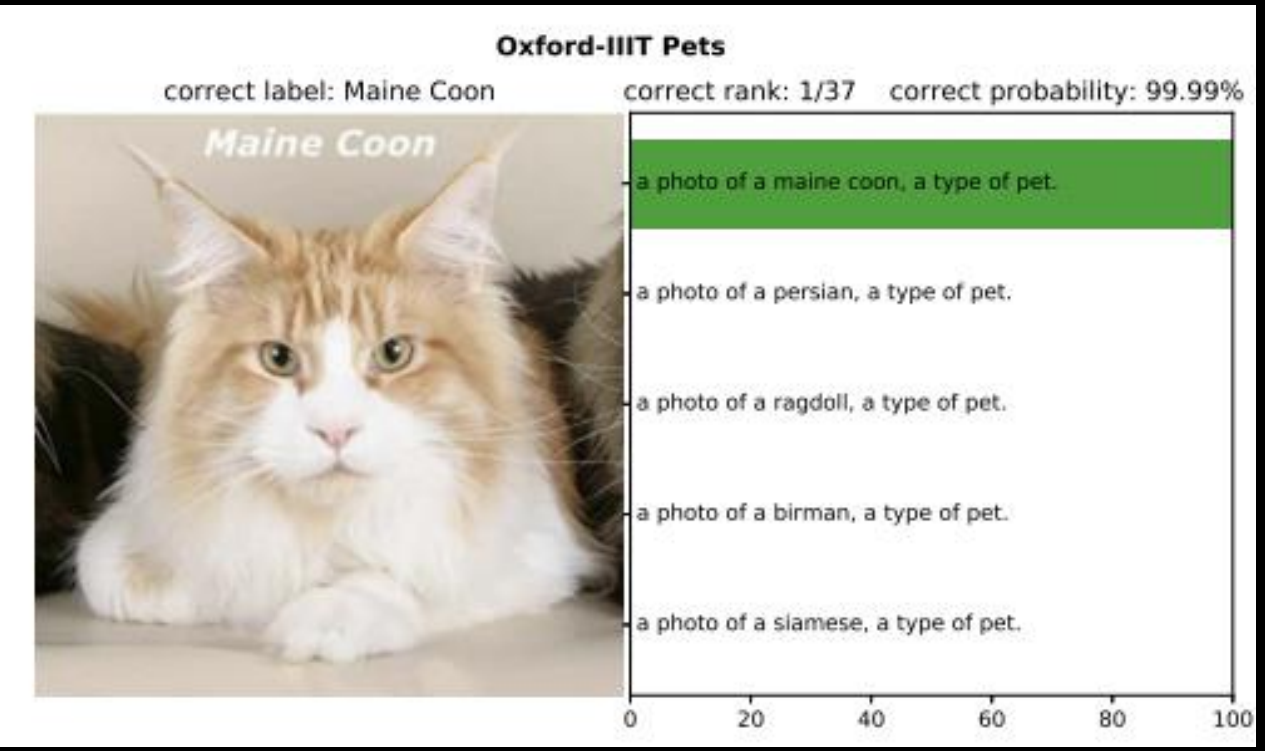
e.g., zero-shot classification:

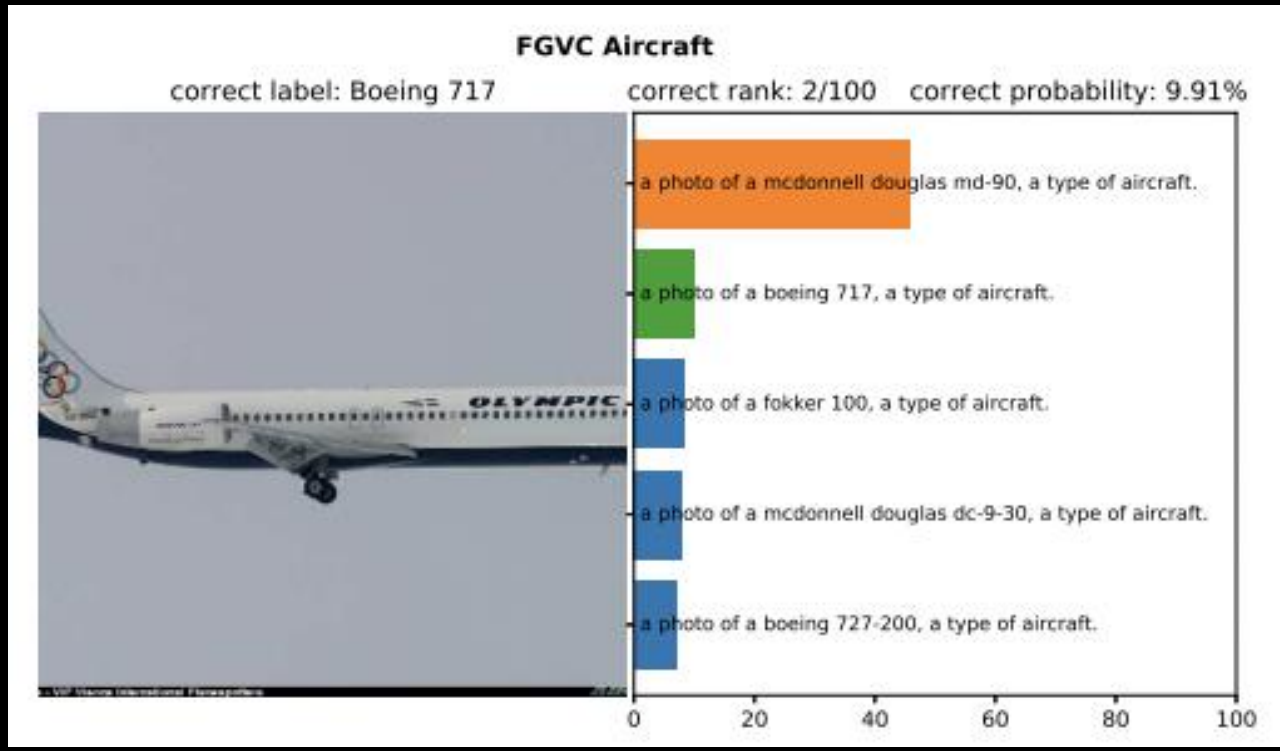
1. Encode features for all candidate classes
2. Encode feature for image
3. Compute cosine similarity between features for image and all text candidates
4. Apply softmax to identify most probable match (i.e., highest score)



Radford et al. Learning Transferable Visual Models From Natural Language Supervision. ICML 2021

Fine-Grained Classification Predictions; e.g.,





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CLIP Benchmark Datasets

Subset of datasets shown here:

Classification evaluation spanned fine-grained classification (e.g., food, bird, aircraft, and car categories), distribution shifts for ImageNet categories (e.g., corrupted images), and more

Dataset	Classes	Train size	Test size	Evaluation metric
Food-101	102	75,750	25,250	accuracy
CIFAR-10	10	50,000	10,000	accuracy
CIFAR-100	100	50,000	10,000	accuracy
Birdsnap	500	42,283	2,149	accuracy
SUN397	397	19,850	19,850	accuracy
Stanford Cars	196	8,144	8,041	accuracy
FGVC Aircraft	100	6,667	3,333	mean per class
Pascal VOC 2007 Classification	20	5,011	4,952	11-point mAP
Describable Textures	47	3,760	1,880	accuracy
Oxford-IIIT Pets	37	3,680	3,669	mean per class
Caltech-101	102	3,060	6,085	mean-per-class
Oxford Flowers 102	102	2,040	6,149	mean per class
MNIST	10	60,000	10,000	accuracy
Facial Emotion Recognition 2013	8	32,140	3,574	accuracy
STL-10	10	1000	8000	accuracy
EuroSAT	10	10,000	5,000	accuracy
RESISC45	45	3,150	25,200	accuracy
GTSRB	43	26,640	12,630	accuracy
KITTI	4	6,770	711	accuracy
Country211	211	43,200	21,100	accuracy
PatchCamelyon	2	294,912	32,768	accuracy
UCF101	101	9,537	1,794	accuracy
Kinetics700	700	494,801	31,669	mean(top1, top5)
CLEVR Counts	8	2,000	500	accuracy
Hateful Memes	2	8,500	500	ROC AUC
Rendered SST2	2	7,792	1,821	accuracy
ImageNet	1000	1,281,167	50,000	accuracy

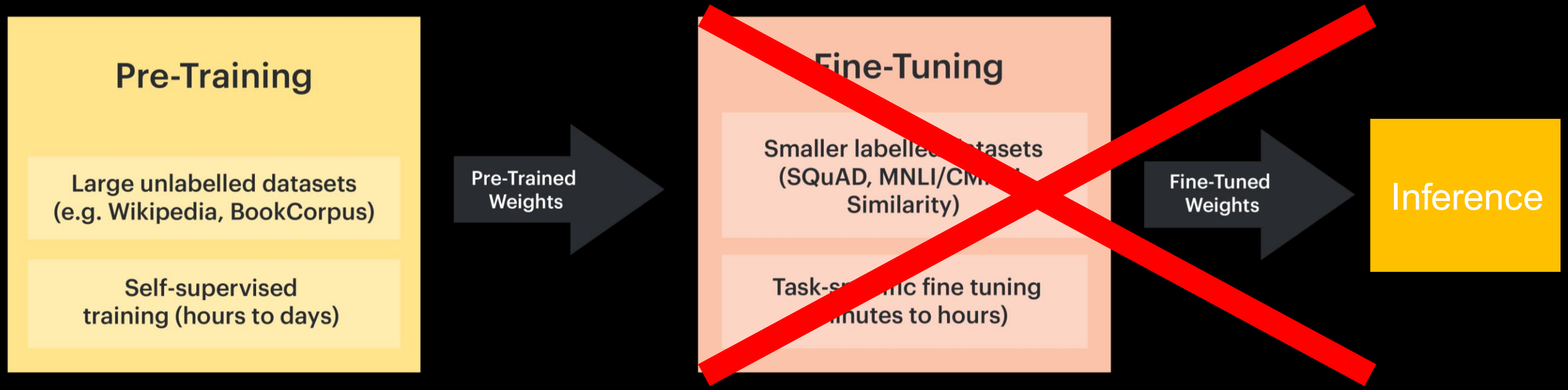
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The Internet offered offered a suitable multi-task learning environment, inherently representing a wide range of domains and tasks, for learning such representations.

The vision from 2007 paper (from Bengio and LeCun) now became a reality:

"... one should not consider the large variety of [AI] tasks separately, but as different aspects of a more general problem: that of learning the basic structure of the world, as seen say through the eyes and ears of a growing animal or a young child. This is an instance of multi-task learning... after training such a system on a large variety of tasks in the AI-set, the system may generalize to a new task from only a few labeled examples. We hypothesize that many tasks in the AI-set may be built around common representations..."

Summary: A New Few-Shot/Zero-Shot Paradigm



Across GPT-2, GPT-3, and CLIP, massive models (e.g., ~0.5B to 175B parameters) plus internet-scale training data enabled few-shot and zero-shot learning!

https://docs.graphcore.ai/projects/bert-training/en/latest/bert.html

Today's Topics

- The quest for growth: the secret sauce for success
- Scaling laws: predictable gains from model size, data, & compute
- Fueling scale: harnessing the Internet as training data
- Revelation: few-shot and zero-shot abilities emerge
- Programming tutorial

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The End